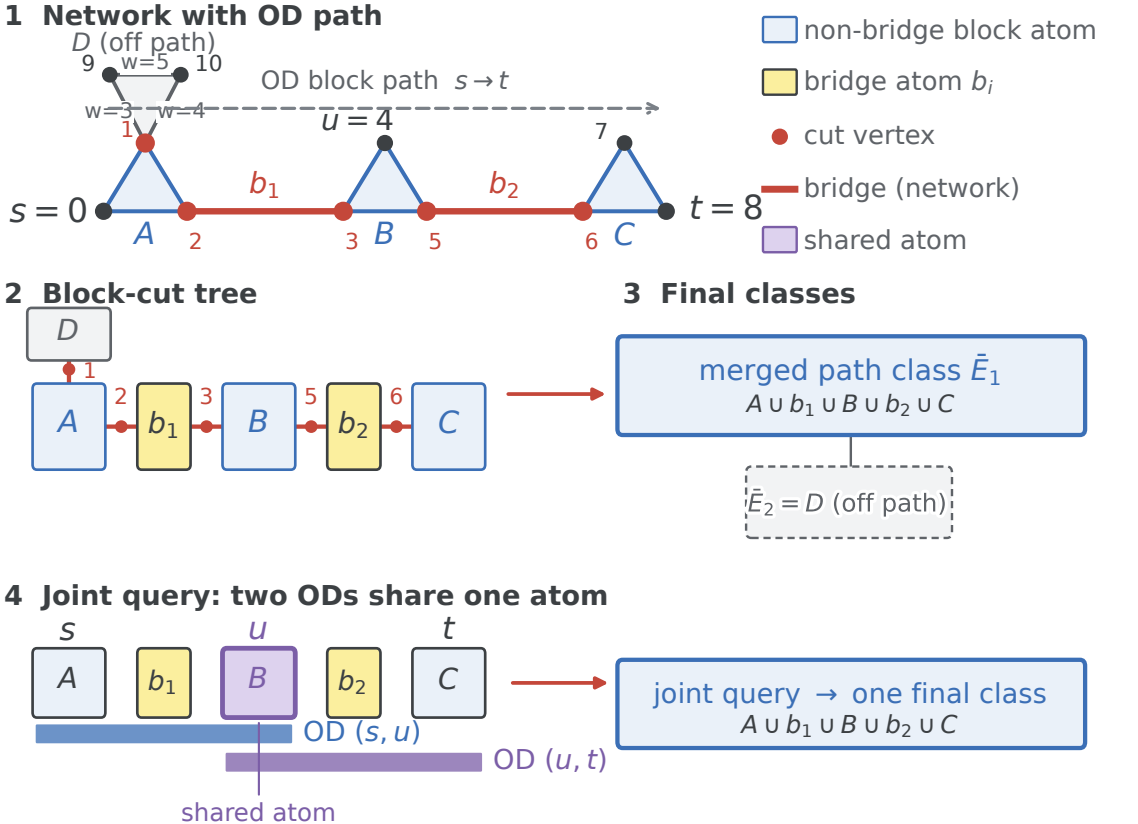
1 Network with OD path
D (off path)
9
w=5
10
w=3
w=4
1
OD block path $s \to t$
$u = 4$
7
$s = 0$
b_1
b_2
$t = 8$
A
2
3
B
5
6
C
non-bridge block atom
bridge atom b_i
cut vertex
bridge (network)
shared atom
2 Block-cut tree
D
1
A
2
b_1
3
B
5
b_2
6
C
3 Final classes
merged path class $\bar{E}_1$
$A \cup b_1 \cup B \cup b_2 \cup C$
$\bar{E}_2 = D$ (off path)
4 Joint query: two ODs share one atom
s
u
t
A
b_1
B
b_2
C
OD (s, u)
OD (u, t)
shared atom
joint query → one final class
$A \cup b_1 \cup B \cup b_2 \cup C$
Region-family contrast: off-path D weights $3, 4, 5$
All regions: incompatible | Two-sided connected regions: unrestricted off-path weights